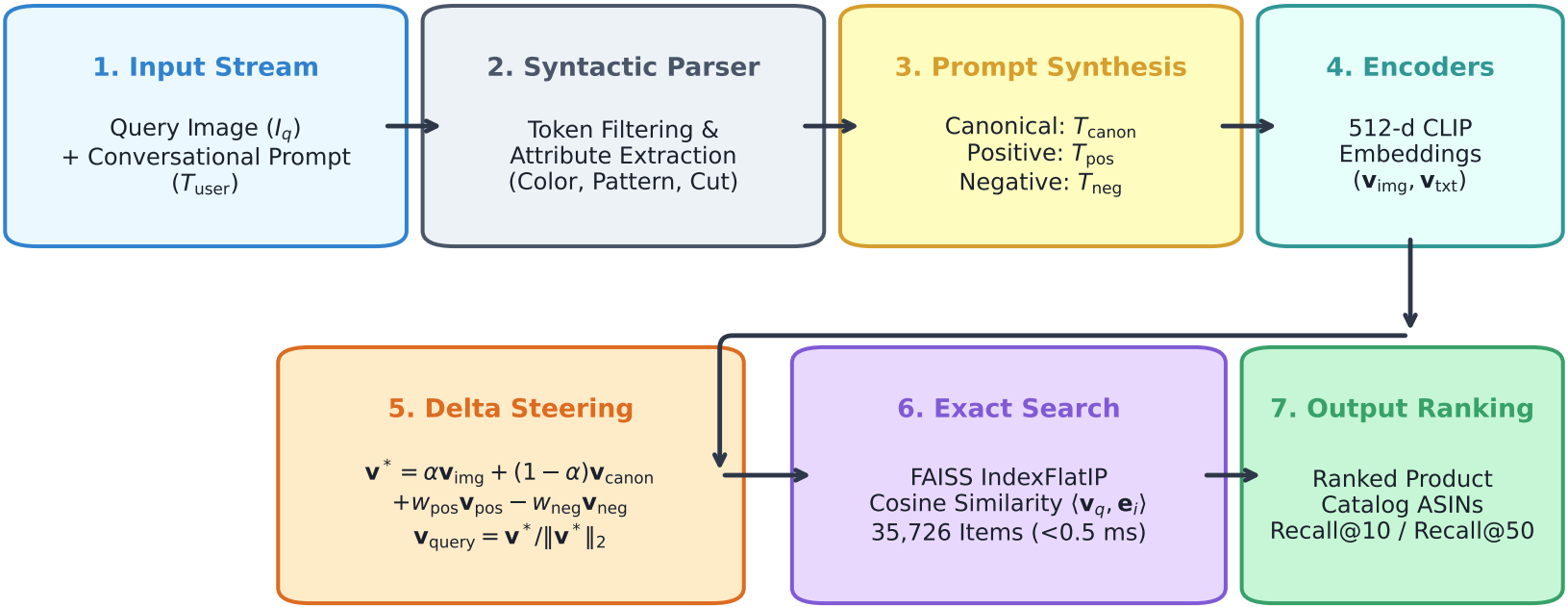
1. Input Stream
Query Image (I_q)
+ Conversational Prompt
(T_{user})
2. Syntactic Parser
Token Filtering &
Attribute Extraction
(Color, Pattern, Cut)
3. Prompt Synthesis
Canonical: T_{canon}
Positive: T_{pos}
Negative: T_{neg}
4. Encoders
512-d CLIP
Embeddings
($\mathbf{v}_{\text{img}}, \mathbf{v}_{\text{txt}}$)
5. Delta Steering
$\mathbf{v}^* = \alpha \mathbf{v}_{\text{img}} + (1-\alpha)\mathbf{v}_{\text{canon}}$
$+ w_{\text{pos}}\mathbf{v}_{\text{pos}} - w_{\text{neg}}\mathbf{v}_{\text{neg}}$
$\mathbf{v}_{\text{query}} = \mathbf{v}^* / \|\mathbf{v}^*\|_2$
6. Exact Search
FAISS IndexFlatIP
Cosine Similarity $\langle \mathbf{v}_q, \mathbf{e}_i \rangle$
35,726 Items (<0.5 ms)
7. Output Ranking
Ranked Product
Catalog ASINs
Recall@10 / Recall@50